

4th Semester ES-DS

Syllabus

Unit-1 Air pollution and its control techniques:

Contaminant behavior in the environment, Air pollution due to SO_x, NO_x, photochemical smog. Indoor air pollution, Natural pathways for degradation: Carbon cycle, Sulphur cycle, Nitrogen cycle, Oxygen cycle. Factors responsible for altering the composition of atmosphere (deforestation, burning of fossil fuels, industrial and vehicular emissions, CFCs). Techniques to control Air pollution, ambient air quality and continuous air quality monitoring, Control measures at source, Kyoto Protocol, Carbon Credits.

Unit -2 Water pollution and its control techniques:

Major sources of water pollution: Eutrophication, acid mine drains, pesticides and fertilizers, dyeing and tanning, marine pollution, micro plastics, Techniques to control water pollution: Conventional waste water treatment-types of sewage, sewerage system, alternative systems, primary, secondary and tertiary processes including aerobic and anaerobic techniques, safe disposal and its utility. Treatment schemes for waste water from dairy, textile, power plants, pharmaceutical industries, and agro based industries such as rice mills

Unit -3 Other Environmental Pollution & Waste Management:

Soil pollution: Soil around us, Soil water characteristics, soil pollution. Causes, effects & control: noise pollution, nuclear & radiation hazards, marine pollution (Oil spills & Ocean Acidification)

Solid waste management: Composting, Vermiculture, landfills, hazardous waste treatment, bioremediation technologies, conventional techniques (land farming, constructed wetlands), and phytoremediation. Degradation of xenobiotics in environment: Petroleum hydrocarbons, pesticides, heavy metals Introduction, types of e-wastes, environmental impact, e-waste recycling, e-waste management rules.

Unit -4 Social Issues and the Environmental Laws

Concept of Sustainable development, Water conservation, rain water harvesting, watershed Management, Resettlement and rehabilitation of people; its problems and concerns. Environmental Laws (brief idea only)

Environment Protection Act, Air (Prevention and Control of Pollution) Act, Water (Prevention and control of Pollution) Act, Wildlife Protection Act, Forest Conservation Act Issues involved in enforcement of environmental legislation. Different government initiatives (brief idea only)- National ambient air quality standard 2009, Swachh Bharat Abhiyan, National afforestation program and Act-2016, National River conservation plan and National Ganga River basin authority, Formation of National Green Tribunal

UNIT-1: AIR POLLUTION AND ITS CONTROL TECHNIQUES

Introduction to Air Pollution

Air pollution refers to the presence of harmful substances (solid, liquid, or gaseous) in the atmosphere in concentrations that are injurious to human health, living organisms, and the environment. These substances, known as air pollutants, disturb the natural composition of air and lead to environmental degradation.

Contaminant Behaviour in the Environment

Once pollutants are released into the atmosphere, they do not remain stationary. Their behaviour depends on various physical and chemical processes that determine their transport, transformation, and removal.

Transport and Transformation of Air Pollutants

Air pollutants released into the atmosphere from sources such as industries, vehicles, and power plants do not remain confined to their point of origin. Instead, they undergo continuous movement and chemical changes in the atmosphere. These processes are broadly classified as transport and transformation of pollutants. Together, they determine the distribution, concentration, and impact of pollutants on the environment and human health.

Transport of Pollutants

Transport refers to the movement of pollutants from one place to another in the atmosphere. This movement is governed by various physical processes that influence how pollutants spread over local, regional, or even global scales.

One of the most important mechanisms is wind movement, also known as advection. In this process, air currents carry pollutants horizontally across large distances. Strong winds can transport pollutants from industrial areas to distant regions, sometimes even across countries. While this helps in diluting pollutants at the source, it increases the geographical area affected by pollution.

Another important process is diffusion, which involves the movement of pollutants from regions of higher concentration to regions of lower concentration. This occurs due to random molecular motion and results in gradual mixing of pollutants with the surrounding air. Diffusion plays a significant role at smaller scales and helps in reducing localized pollution levels.

In addition to advection and diffusion, turbulence plays a crucial role in pollutant transport. Turbulence is the irregular and chaotic movement of air caused by atmospheric instability, temperature differences, and obstacles such as buildings or terrain. It promotes both horizontal and vertical mixing of pollutants, thereby reducing their concentration near the ground level and dispersing them into the atmosphere.

Through these mechanisms, pollutants can spread widely, affecting air quality over large regions and influencing environmental conditions far from the original source.

Transformation of Pollutants

After being transported in the atmosphere, pollutants often undergo chemical and photochemical reactions, leading to the formation of new substances. This process is known as transformation.

Pollutants are classified into primary and secondary pollutants based on their origin. Primary pollutants are those that are directly emitted into the atmosphere from sources such as vehicles, industries, and power plants. Examples include sulfur dioxide (SO_2), nitrogen oxides (NO_x), and carbon monoxide (CO).

Secondary pollutants, on the other hand, are not emitted directly. They are formed in the atmosphere through chemical reactions involving primary pollutants. These reactions are often driven by sunlight, moisture, and atmospheric conditions, and the resulting secondary pollutants are usually more harmful than the original substances.

One important example of transformation is the formation of acid rain. Sulfur dioxide released from burning fossil fuels reacts with oxygen to form sulfur trioxide, which further reacts with water vapour to produce sulfuric acid. This acid mixes with rainwater and falls to the ground, causing damage to soil, vegetation, aquatic life, and buildings.

Another significant transformation process is the formation of photochemical smog. Nitrogen oxides react with hydrocarbons in the presence of sunlight to produce ozone and other oxidizing agents such as peroxyacetyl nitrate (PAN). These substances form a brownish haze known as smog, which is harmful to human health, reduces visibility, and damages crops.

Environmental Significance

The combined processes of transport and transformation are crucial in determining the overall impact of air pollution. Transport spreads pollutants over wide areas, increasing exposure levels, while transformation often increases their toxicity by converting them into more harmful secondary pollutants.

These processes:

- Extend pollution beyond the source region
- Increase environmental and health risks
- Complicate pollution control strategies
- Contribute to global issues like acid rain and smog

Air Pollution due to Sulfur Oxides (SO_x)

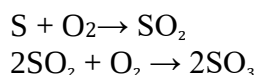
Sulfur oxides, primarily sulfur dioxide (SO₂) and sulfur trioxide (SO₃), are major air pollutants generated from the burning of sulfur-containing fuels like coal and petroleum.

Sources of SO_x

SO_x emissions originate from both natural and human activities. Natural sources include volcanic eruptions and emissions from oceans, whereas anthropogenic sources are more significant and include coal-based thermal power plants, petroleum refineries, and various industrial processes.

Formation of SO_x

Sulfur present in fuels reacts with oxygen during combustion to form sulfur dioxide, which can further oxidize into sulfur trioxide:



Effects of SO_x

1. Human Health Effects

Exposure to sulfur oxides causes irritation of the respiratory tract and aggravates diseases such as asthma and bronchitis. Long-term exposure can lead to severe lung damage and reduced lung function.

2. Environmental Effects

SO_x plays a major role in the formation of acid rain. Sulfur dioxide reacts with oxygen and water vapour to form sulfuric acid (H₂SO₄), which falls with rainwater. Acid rain leads to:

- Damage to crops and forests
- Loss of soil fertility due to nutrient leaching
- Harm to aquatic ecosystems

3. Material Damage

Sulfur oxides cause corrosion of metals and deterioration of buildings, monuments, and historical structures by reacting with materials and forming acids.

Summary of Effects of SO_x

Area	Impact
Human health	Respiratory diseases
Environment	Acid rain, ecosystem damage
Materials	Corrosion and structural damage

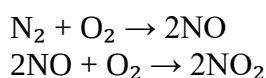
Air Pollution due to Nitrogen Oxides (NOx)

Nitrogen oxides mainly include nitric oxide (NO) and nitrogen dioxide (NO₂). These gases are formed during high-temperature combustion processes where nitrogen and oxygen in the air react.

Sources of NOx

Natural sources include lightning and microbial activity in soil, while major anthropogenic sources include automobiles, thermal power plants, and industrial boilers.

Formation of NOx



Effects of NOx

1. Human Health Effects

NOx gases irritate the lungs, reduce immunity, and cause breathing difficulties. Prolonged exposure can lead to chronic respiratory diseases.

2. Environmental Effects

One of the most significant impacts of NOx is the formation of photochemical smog. Nitrogen oxides react with hydrocarbons in the presence of sunlight to produce ozone (O₃) and peroxyacetyl nitrate (PAN), forming a brownish haze.

Additionally, NO₂ reacts with water vapor to form nitric acid (HNO₃), contributing to acid rain.

3. Visibility Reduction

NOx contributes to haze formation, which reduces visibility and affects transportation safety and overall air quality.

Summary of Effects of NOx

Area	Impact
Health	Lung irritation and diseases
Environment	Smog and acid rain
Visibility	Haze and reduced clarity

Photochemical Smog

Photochemical smog is a type of air pollution formed by the reaction of sunlight with primary pollutants such as nitrogen oxides (NOx) and hydrocarbons (volatile organic compounds – VOCs). It is commonly observed in urban areas with heavy traffic and industrial activities, especially under conditions of strong sunlight and low wind.

It appears as a brownish or yellowish haze in the atmosphere and is highly harmful to humans, plants, and materials.

Formation of Photochemical Smog (Mechanism)

The formation of photochemical smog is a complex process involving a series of photochemical reactions:

1. Emission of Primary Pollutants

- Nitrogen oxides (NO, NO₂) from vehicles
- Hydrocarbons (VOCs) from fuel evaporation and industries

2. Action of Sunlight

- Sunlight provides energy for chemical reactions.

3. Photodissociation of NO₂

- $\text{NO}_2 \rightarrow \text{NO} + \text{O}$ (in presence of sunlight)

4. Formation of Ozone (O₃)

- $\text{O} + \text{O}_2 \rightarrow \text{O}_3$

5. Formation of Secondary Pollutants

- Ozone (O₃)
- Peroxyacetyl nitrate (PAN)
- Aldehydes

These secondary pollutants together form photochemical smog.

Effects of Photochemical Smog

Photochemical smog has a wide range of harmful effects on human health, vegetation, materials, and the environment. In terms of human health, it causes irritation of the eyes, nose, and throat, along with symptoms such as headache, dizziness, and breathing difficulties. Prolonged exposure can aggravate respiratory diseases like asthma and may lead to reduced lung function. Photochemical smog also has serious effects on vegetation, as it damages leaf tissues, interferes with the process of photosynthesis, and ultimately results in a decrease in crop yield. In addition to its impact on living organisms, it affects materials by causing cracking of rubber and deterioration of paints and fabrics. Furthermore, photochemical smog contributes to environmental degradation by reducing visibility and forming ground-level ozone, which is a harmful pollutant that adversely affects both human health and ecosystems.

Indoor Air Pollution

Indoor air pollution refers to the contamination of air inside buildings such as homes, offices, schools, and industries. Since people spend a large portion of their time indoors, indoor air pollution poses a significant risk to human health.

Sources of Indoor Air Pollution

Indoor air pollution arises from multiple sources, which can be broadly classified as follows:

1. Combustion Sources

Combustion activities are one of the major contributors to indoor pollution. The burning of fuels such as wood, coal, and kerosene for cooking releases harmful gases and particulate matter. In addition, cigarette smoke and emissions from gas stoves and heaters introduce toxic pollutants like carbon monoxide into indoor environments.

2. Building Materials

Many construction and furnishing materials emit harmful substances over time. Paints and varnishes release volatile organic compounds (VOCs), while materials like asbestos can release hazardous fibers. Furniture and pressed wood products often emit formaldehyde, which can negatively affect indoor air quality.

3. Biological Sources

Biological contaminants also play a significant role in indoor air pollution. Mold and fungi grow in damp conditions, while bacteria and viruses spread through air. Dust mites and pet dander are common allergens that can trigger respiratory issues and allergic reactions.

4. Chemical Sources

Various household chemicals contribute to indoor pollution. Cleaning agents, pesticides, and air fresheners release chemical vapours that can accumulate in enclosed spaces and affect human health.

Effects of Indoor Air Pollution

The effects of indoor air pollution can be categorized into short-term and long-term impacts depending on the duration and level of exposure.

1. Short-Term Effects

Short-term exposure to indoor pollutants can lead to immediate health problems such as irritation of the eyes, nose, and throat. Individuals may experience headaches, dizziness, and allergic reactions, which often disappear once the exposure is reduced.

2. Long-Term Effects

Prolonged exposure to indoor air pollutants can cause serious health conditions. These include chronic respiratory diseases, lung cancer, heart diseases, and a general weakening of the immune system. Long-term exposure is particularly dangerous because its effects develop gradually and may not be immediately noticeable.

Vulnerable Groups

Certain groups of people are more sensitive to indoor air pollution due to their physiological conditions. These include:

- Children, whose lungs are still developing
- Elderly individuals with weaker immune systems
- Pregnant women, where exposure may affect fetal development
- People suffering from respiratory diseases such as asthma

Indoor Pollutants and Their Effects

Pollutant	Source	Effect
Carbon Monoxide (CO)	Incomplete combustion	Suffocation, reduced oxygen supply
Volatile Organic Compounds (VOCs)	Paints, cleaners	Headache, irritation
Particulate Matter	Smoke, dust	Lung diseases
Biological Pollutants	Mold, bacteria	Allergies, infections

Natural Pathways for Degradation (Biogeochemical Cycles)

Introduction

In nature, various elements are continuously recycled through different components of the environment such as the atmosphere, hydrosphere, lithosphere, and biosphere. These natural recycling processes are known as biogeochemical cycles. They play a vital role in maintaining ecological balance by ensuring that essential elements are reused and not permanently lost. The major natural pathways for degradation include the carbon cycle, sulfur cycle, nitrogen cycle, and oxygen cycle.

1. Carbon Cycle

Description

The carbon cycle is the process by which carbon is circulated between the atmosphere, living organisms, soil, and oceans.

Process

- Carbon dioxide (CO_2) is absorbed by plants during photosynthesis
- Plants convert CO_2 into organic compounds
- Animals consume plants and obtain carbon
- Carbon returns to the atmosphere through respiration
- Decomposition of dead organisms releases CO_2
- Burning of fossil fuels also adds CO_2 to the atmosphere

Importance

- Maintains atmospheric CO_2 balance
- Supports life processes like photosynthesis and respiration
- Regulates global climate

2. Sulphur Cycle

Description

The sulphur cycle involves the movement of sulfur through air, water, soil, and living organisms.

Process

- Sulfur is released into the atmosphere from volcanic eruptions and decomposition
- It exists in the form of sulfur dioxide (SO_2)
- SO_2 reacts with water to form sulfuric acid
- Returns to earth as acid rain
- Plants absorb sulfur from soil
- Animals obtain sulfur through food
- Decomposition releases sulfur back to the environment

Importance

- Essential for protein synthesis in organisms
- Maintains soil fertility
- Links atmosphere with land and water systems

3. Nitrogen Cycle

Description

The nitrogen cycle is the process by which nitrogen is converted into different chemical forms and circulated through the environment.

Process

1. Nitrogen Fixation

- Atmospheric nitrogen (N_2) is converted into ammonia by bacteria

2. Nitrification

- Ammonia is converted into nitrites and nitrates

3. Assimilation

- Plants absorb nitrates and use them for growth

4. Ammonification

- Decomposition of organic matter releases ammonia

5. Denitrification

- Nitrates are converted back into nitrogen gas and released into the atmosphere

Importance

- Essential for formation of proteins and DNA
- Maintains soil fertility
- Supports plant growth

4. Oxygen Cycle

Description

The oxygen cycle describes the movement of oxygen within the atmosphere, biosphere, and lithosphere.

Process

- Plants release oxygen during photosynthesis
- Animals and humans consume oxygen during respiration
- Oxygen is used in combustion processes
- Decomposition also consumes oxygen
- Oxygen is continuously recycled between organisms and the atmosphere

Importance

- Essential for respiration of living organisms
- Supports combustion processes
- Maintains atmospheric balance

Summary Table

Cycle	Main Process	Importance
Carbon Cycle	Photosynthesis & respiration	Climate regulation
Sulphur Cycle	Atmospheric & soil circulation	Soil fertility
Nitrogen Cycle	Fixation & nitrification	Plant growth
Oxygen Cycle	Photosynthesis & respiration	Life support

Deforestation is one of the major factors responsible for altering the composition of the atmosphere. It refers to the large-scale cutting of forests for agriculture, urbanization, and industrial development. Forests act as carbon sinks, absorbing carbon dioxide (CO₂) and releasing oxygen (O₂). However, when trees are removed, the absorption of CO₂ decreases, oxygen production reduces, and the stored carbon is released back into the atmosphere, leading to serious environmental issues such as global warming and ecological imbalance.

Key impacts of deforestation:

- Decrease in CO₂ absorption
- Reduction in oxygen levels
- Increase in greenhouse gases
- Loss of biodiversity
- Disturbance in ecological balance

The burning of fossil fuels such as coal, petroleum, and natural gas is another major contributor to atmospheric changes. These fuels are extensively used in energy generation, transportation, and industrial activities. Their combustion releases harmful pollutants like CO₂, SO₂, NO_x, and particulate matter, which significantly affect atmospheric composition. These emissions enhance the greenhouse effect, leading to climate change and global warming, and also contribute to acid rain formation.

Key impacts of fossil fuel burning:

- Release of greenhouse gases (CO₂)
- Formation of acid rain (SO₂, NO_x)
- Air pollution and reduced air quality
- Adverse health effects on humans
- Contribution to climate change

Industrial and vehicular emissions play a significant role in deteriorating air quality. Rapid industrialization and the increasing number of vehicles have led to higher emissions of pollutants such as carbon monoxide (CO), nitrogen oxides (NO_x), hydrocarbons (VOCs), and sulfur oxides (SO_x). These pollutants contribute to the formation of photochemical smog,

reduce visibility, and degrade environmental quality. Continuous exposure to such pollutants can lead to serious health issues.

Key impacts of industrial and vehicular emissions:

- Formation of photochemical smog
- Reduction in air quality and visibility
- Respiratory and cardiovascular diseases
- Environmental degradation
- Increased pollution levels in urban areas

Chlorofluorocarbons (CFCs) are synthetic chemical compounds used in refrigeration, air conditioning, and aerosol sprays. Although they are stable in the lower atmosphere, they break down in the stratosphere under ultraviolet radiation, releasing chlorine atoms that destroy ozone molecules. This leads to ozone layer depletion, which increases the penetration of harmful ultraviolet (UV) radiation to the Earth's surface.

Key impacts of CFCs:

- Depletion of ozone layer
- Increase in harmful UV radiation
- Risk of skin cancer and eye diseases
- Damage to crops and marine ecosystems
- Long-term environmental imbalance

Air Pollution Control Techniques, Air Quality Monitoring, Kyoto Protocol & Carbon Credits

1. Techniques to Control Air Pollution

Air pollution control involves a combination of preventive and corrective measures aimed at reducing the emission of harmful pollutants into the atmosphere. These techniques focus on minimizing pollution at its source, treating pollutants before release, and improving environmental conditions through sustainable practices. Effective control requires technological solutions, regulatory policies, and public awareness.

Major air pollution control techniques:

- Use of clean and renewable energy sources (solar, wind, CNG)
- Installation of pollution control devices like electrostatic precipitators, scrubbers, and filters
- Adoption of energy-efficient technologies
- Promotion of public transport and electric vehicles
- Afforestation and green belts to absorb pollutants

2. Ambient Air Quality and Continuous Monitoring

Ambient air quality refers to the quality of outdoor air in a particular area, measured in terms of pollutant concentration levels. Monitoring air quality is essential to assess pollution levels, ensure compliance with standards, and protect public health. Continuous air quality

monitoring systems provide real-time data on pollutants such as particulate matter (PM_{2.5}, PM₁₀), SO₂, NO_x, CO, and ozone.

Key aspects of air quality monitoring:

- Measurement of major pollutants (PM, SO₂, NO_x, CO, O₃)
- Use of automatic monitoring stations
- Real-time data collection and analysis
- Comparison with air quality standards
- Public awareness through Air Quality Index (AQI)

3. Control Measures at Source

Controlling pollution at the source is the most effective way to reduce air pollution. It involves modifying processes, fuels, and equipment to minimize pollutant generation before they enter the atmosphere.

Important source control measures:

- Use of low-sulfur and clean fuels
- Proper maintenance of vehicles and machinery
- Installation of catalytic converters in vehicles
- Process modification in industries
- Adoption of low-emission technologies

4. Kyoto Protocol

The Kyoto Protocol is an international treaty aimed at reducing greenhouse gas emissions to combat global warming. It was adopted in 1997 and came into force in 2005 under the United Nations Framework Convention on Climate Change (UNFCCC).

The protocol sets legally binding emission reduction targets for developed countries and promotes sustainable development.

Key features of the Kyoto Protocol:

- Reduction of greenhouse gas emissions
- Binding targets for developed countries
- Promotion of clean development mechanisms (CDM)
- Encouragement of international cooperation

5. Carbon Credits

Carbon credits are a market-based mechanism developed under the Kyoto Protocol to reduce greenhouse gas emissions. One carbon credit represents the reduction of one tonne of CO₂ or equivalent greenhouse gases.

Organizations that reduce emissions below their limits can sell excess credits, while those exceeding limits must purchase credits.

Key points about carbon credits:

- Encourages reduction of emissions
- Promotes clean and green technologies
- Supports sustainable development projects
- Creates an economic incentive for pollution control.

UNIT-2: Water Pollution and Its Control Techniques

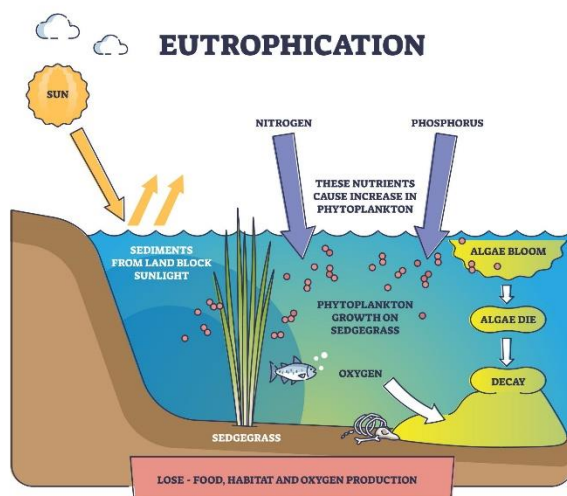
Introduction

Water pollution refers to the contamination of water bodies such as rivers, lakes, oceans, and groundwater due to the discharge of harmful substances. It affects aquatic life, human health, and overall ecosystem balance. Rapid industrialization, urbanization, and agricultural activities are the major contributors to water pollution. Effective control techniques and proper treatment systems are essential to maintain water quality and ensure sustainable use.

Major Sources of Water Pollution

1. Eutrophication

Eutrophication is a process in which water bodies become enriched with nutrients such as nitrogen and phosphorus, mainly due to the discharge of fertilizers and sewage. These excess nutrients promote the rapid growth of algae, forming algal blooms that cover the water surface. This blocks sunlight penetration and reduces the dissolved oxygen content in water, creating hypoxic conditions. As a result, aquatic organisms such as fish and other aerobic species are unable to survive, leading to ecological imbalance and deterioration of water quality.



Key points:

- Caused by fertilizer runoff and sewage discharge
- Leads to excessive algal growth (algal blooms)
- Causes oxygen depletion (hypoxia)
- Results in death of aquatic organisms
- Produces foul smell and degraded water quality

2. Acid Mine Drainage (AMD)

Acid mine drainage is a serious form of water pollution associated with mining activities. It occurs when sulfide minerals present in exposed rocks react with oxygen and water to form sulfuric acid. This acidic water further dissolves heavy metals such as iron, lead, and mercury, which then enter nearby water bodies. The resulting contamination significantly lowers the pH of water and makes it toxic for aquatic life as well as unsafe for human consumption.

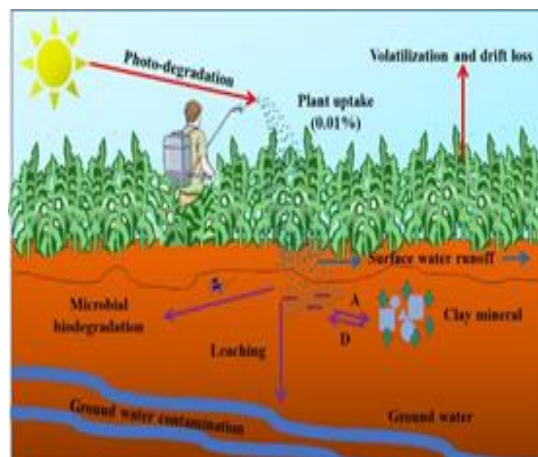
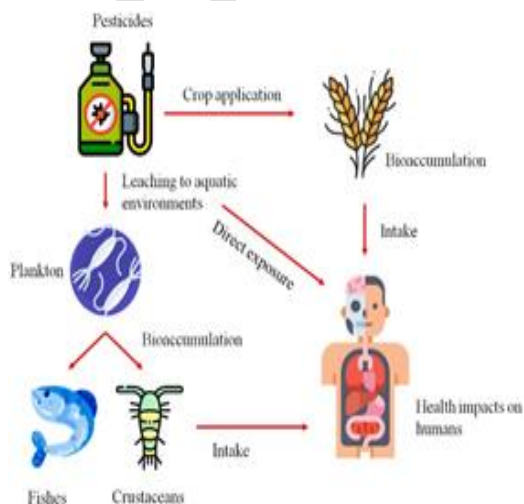
Key points:

- Originates from mining and excavation activities
- Produces acidic water (low pH)
- Releases toxic heavy metals
- Contaminates rivers and groundwater
- Harms aquatic life and drinking water sources



3. Pesticides and Fertilizers

Modern agricultural practices involve extensive use of pesticides and chemical fertilizers to increase crop productivity. However, during rainfall or irrigation, these chemicals are washed into nearby water bodies through surface runoff. These substances introduce toxic compounds into water, affecting aquatic organisms and entering the food chain. Long-term exposure leads to bioaccumulation and biomagnification, posing serious risks to human and animal health.



Key points:

- Introduces toxic chemicals and nutrients into water
- Causes bioaccumulation in food chains
- Leads to eutrophication
- Contaminates drinking water sources
- Affects human and animal health

4. Dyeing and Tanning Industries

Industries such as textile dyeing and leather tanning discharge large volumes of wastewater containing dyes, chemicals, and heavy metals like chromium. These effluents are often highly colored and toxic, making them difficult to treat. The presence of such pollutants increases the chemical oxygen demand (COD) of water, reducing oxygen availability. Additionally, the color blocks sunlight penetration, affecting photosynthesis in aquatic plants.

**Key points:**

- High chemical oxygen demand (COD)
- Contains toxic dyes and heavy metals (chromium)
- Reduces sunlight penetration in water
- Affects aquatic plants and organisms
- Causes serious environmental pollution

5. Marine Pollution

Marine pollution refers to the contamination of oceans and seas due to the disposal of industrial waste, oil spills, plastics, and sewage. These pollutants accumulate in marine environments and affect marine biodiversity. Oil spills form a layer on the water surface, preventing oxygen exchange, while plastics harm marine animals through ingestion and entanglement. Pollutants also enter the human food chain through seafood consumption.

Key points:

- Caused by industrial discharge, oil spills, and plastics
- Affects marine biodiversity and ecosystems
- Enters food chain via seafood
- Causes long-term ecological damage
- Difficult to control and clean



6. Microplastics

Microplastics are extremely small plastic particles formed from the breakdown of larger plastic materials due to physical, chemical, and biological processes. These particles are widely present in water bodies and are highly persistent due to their non-biodegradable nature. Aquatic organisms often ingest microplastics, which can accumulate in their bodies and eventually reach humans through the food chain, causing potential health risks.



Key points:

- Found in water, soil, and living organisms
- Non-biodegradable and persistent in nature
- Ingested by fish and aquatic organisms
- Enters human food chain
- Causes potential health hazards

Techniques to Control Water Pollution

1. Conventional Wastewater Treatment

Wastewater treatment is a systematic process used to remove physical, chemical, and biological contaminants from sewage and industrial effluents before their discharge into the environment. It plays a crucial role in protecting water bodies, maintaining public health, and enabling water reuse. The treatment process includes proper collection, transport, and purification of wastewater through various stages.

Types of Sewage

Sewage is broadly classified based on its origin and composition. Domestic sewage consists of wastewater generated from households, including kitchen, bathing, and toilet waste. Industrial sewage contains effluents discharged from factories and industries, often loaded with chemicals and toxic substances. Storm water sewage includes rainwater runoff, which may carry pollutants such as dust, oil, and debris from urban areas.

Key points:

- Domestic sewage → household wastewater
- Industrial sewage → chemical-rich factory effluents
- Storm water → rainwater carrying surface pollutants

Sewerage System

A sewerage system is an organized network of pipes, pumps, and treatment facilities designed to collect and transport sewage from its source to treatment plants. It ensures hygienic disposal of waste and prevents contamination of land and water bodies.

Key points:

- Ensures safe collection and transport of sewage
- Prevents environmental and groundwater contamination
- Essential for urban sanitation and public health

Alternative Systems

In areas where conventional sewerage systems are not feasible, alternative wastewater management systems are used. These systems are cost-effective, eco-friendly, and suitable for rural or low-density regions.

Key points:

- Septic tanks for decentralized treatment
- Biogas plants for energy generation from waste
- Constructed wetlands for natural filtration
- Suitable for rural and remote areas

Stages of Wastewater Treatment

Wastewater treatment is carried out in three main stages, each designed to remove specific types of pollutants.

1. Primary Treatment (Physical Process)

Primary treatment involves the removal of large and suspended solids through physical methods such as screening and sedimentation. This stage reduces the load on subsequent treatment processes.

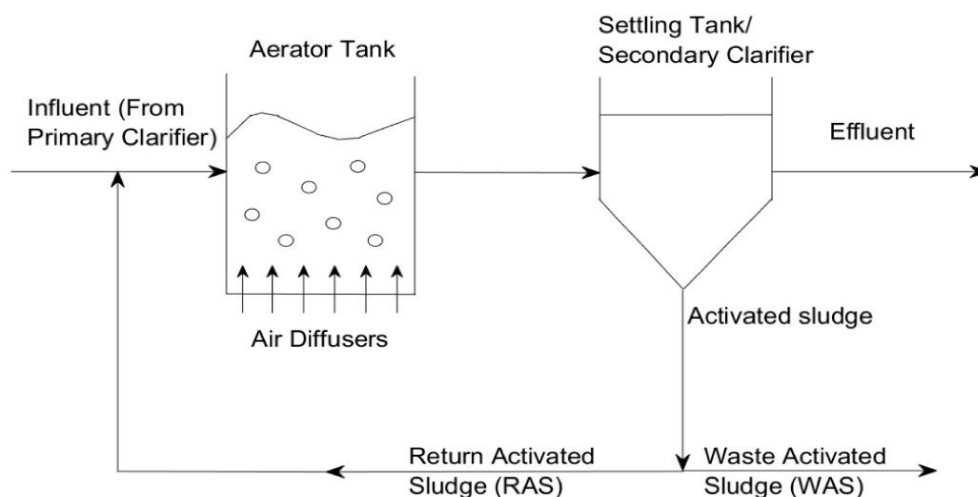


Key points:

- Screening removes large debris
- Sedimentation settles suspended particles
- Reduces turbidity and suspended solids
- Initial step in purification

2. Secondary Treatment (Biological Process)

Secondary treatment uses microorganisms to decompose organic matter present in wastewater. These biological processes convert harmful organic substances into simpler and less harmful compounds.

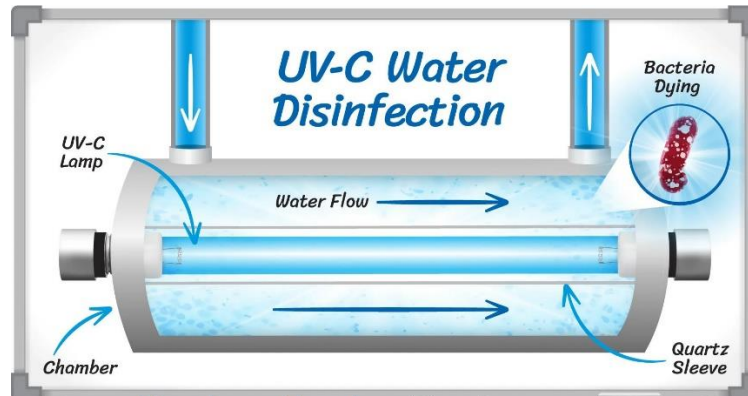


Key points:

- Aerobic processes (activated sludge, trickling filters)
- Anaerobic processes (digestion without oxygen)
- Reduces Biochemical Oxygen Demand (BOD)
- Also lowers Chemical Oxygen Demand (COD)

3. Tertiary Treatment (Advanced Treatment)

Tertiary treatment is the final stage, which further purifies water by removing nutrients, pathogens, and remaining contaminants. It ensures that treated water is safe for reuse or discharge.



Key points:

- Filtration and disinfection (chlorination, UV)
- Removal of nitrogen and phosphorus
- Produces high-quality treated water
- Suitable for reuse and sensitive environments

Safe Disposal and Its Utility

After treatment, wastewater must be safely disposed of or reused to prevent environmental pollution. Treated water can be used for various beneficial purposes, reducing the demand for fresh water resources.

Key points:

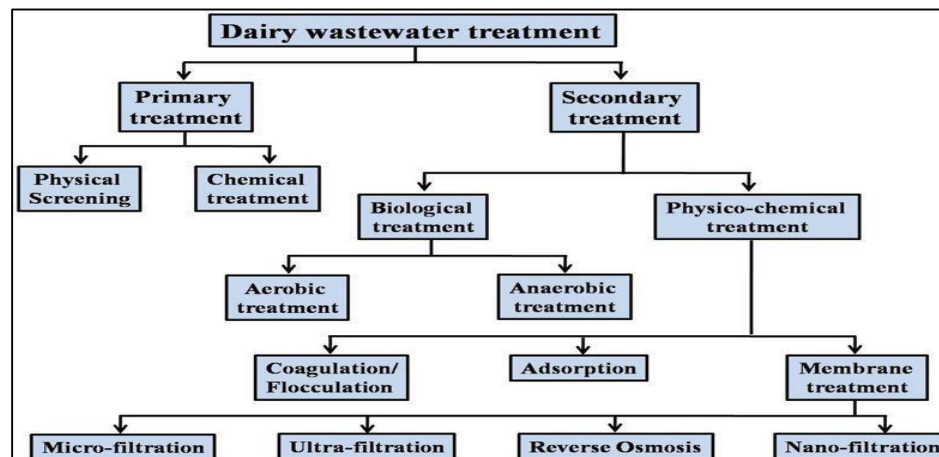
- Used for irrigation in agriculture
- Reused in industrial processes
- Helps in groundwater recharge
- Reduces water scarcity and pollution load

Treatment Schemes for Industrial Wastewater

Industrial wastewater varies significantly depending on the type of industry and the nature of raw materials used. It often contains a combination of organic matter, toxic chemicals, heavy metals, and suspended solids. Therefore, each industry requires specific treatment schemes to effectively remove pollutants and ensure safe discharge or reuse of water. The treatment typically involves a combination of physical, chemical, and biological processes tailored to the characteristics of the wastewater.

1. Dairy Industry

Wastewater generated from dairy industries is rich in organic matter such as fats, proteins, lactose, and suspended solids. These results in high Biochemical Oxygen Demand (BOD) and Chemical Oxygen Demand (COD), making it highly polluting if discharged untreated. The treatment scheme generally begins with screening to remove large particles, followed by biological processes that break down organic matter. Anaerobic digestion is often used, which not only reduces pollution but also produces biogas that can be used as an energy source.

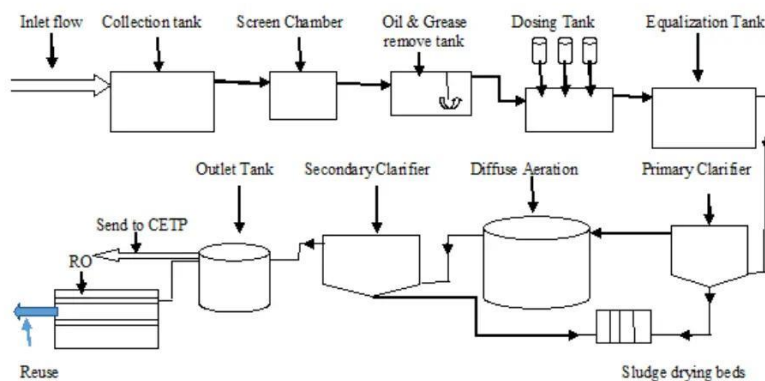


Key points:

- High organic load (BOD & COD) due to fats and proteins
- Initial screening and sedimentation required
- Biological treatment (aerobic/anaerobic) for decomposition
- Anaerobic digestion produces biogas (energy recovery)
- Sludge can be used as manure

2. Textile Industry

Textile industry wastewater contains dyes, chemicals, salts, and sometimes heavy metals, making it highly colored and toxic. The presence of color reduces light penetration in water and affects aquatic life. Treatment schemes focus on color removal, reduction of toxicity, and lowering COD levels. Chemical methods such as coagulation and oxidation are commonly used, along with biological processes. Advanced techniques like adsorption and membrane filtration are also employed for effective purification.



Effluent Treatment for Textile Industries

Key points:

- Contains toxic dyes, chemicals, and salts
- High COD and color intensity
- Color removal is essential before discharge
- Use of chemical (coagulation, oxidation) + biological treatment
- Advanced methods like adsorption, membrane filtration
- Ensures reduction in toxicity and environmental impact

3. Power Plants

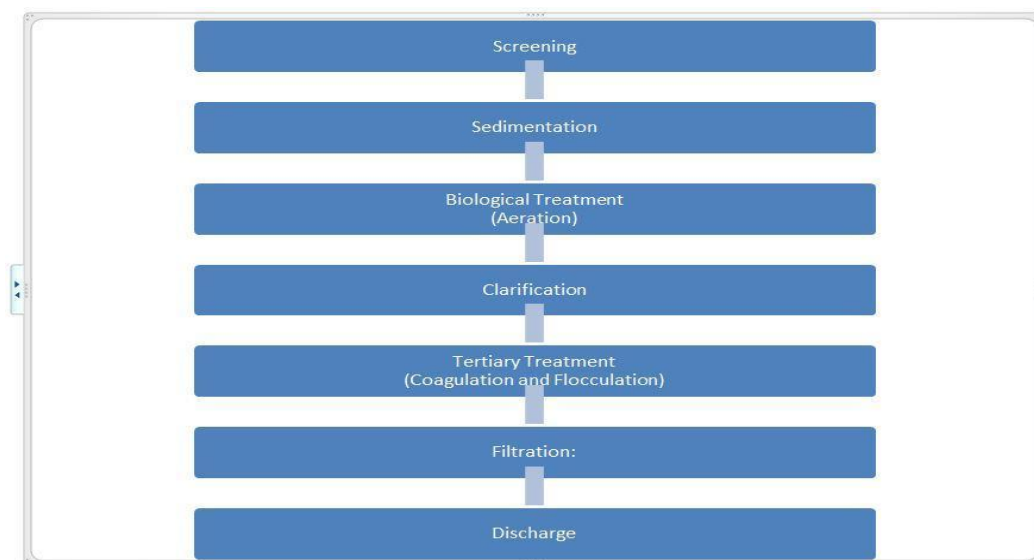
Wastewater from power plants mainly includes cooling water, boiler blow down, and chemical residues. One of the major concerns is thermal pollution, where heated water is discharged into natural water bodies, affecting aquatic ecosystems. Treatment schemes involve cooling processes, removal of heavy metals, and chemical treatment to neutralize harmful substances. Recycling and reuse of water within the plant are also practiced to minimize discharge.

Key points:

- Presence of heated water (thermal pollution)
- Contains chemicals and heavy metals
- Use of cooling towers and heat exchangers
- Removal of toxic substances before discharge
- Emphasis on water recycling and reuse
- Prevents damage to aquatic ecosystems

4. Pharmaceutical Industry

Pharmaceutical wastewater is highly complex, containing organic compounds, antibiotics, solvents, and toxic chemicals. These substances are often resistant to conventional treatment methods, requiring advanced and multi-stage treatment techniques. A combination of biological, chemical, and physical processes is used to degrade and remove contaminants. Strict environmental regulations are followed to ensure safe disposal.

**Key points:**

- Contains toxic and complex organic compounds
- Requires advanced treatment techniques
- Use of biological + chemical + physical processes
- Removal of persistent and hazardous substances
- Strict regulatory standards and monitoring
- Prevents contamination of water resources

5. Agro-Based Industries (Rice Mills)

Wastewater from rice mills and other agro-based industries contains organic matter, husk particles, and suspended solids. Although biodegradable, it has a high pollution load and requires treatment before discharge. Biological treatment methods such as aerobic and anaerobic processes are commonly used to reduce organic content. Treated water can be reused for irrigation, making the process sustainable and cost-effective.

Key points:

- High organic matter and suspended solids
- Wastewater is biodegradable in nature
- Treated using biological treatment methods
- Can be reused for agricultural irrigation
- Low-cost and eco-friendly treatment options
- Supports sustainable water management

Industrial wastewater treatment is essential to prevent environmental pollution and protect water resources. Each industry generates wastewater with unique characteristics, requiring customized treatment schemes. By using appropriate physical, chemical, and biological methods, pollutants can be effectively removed, enabling safe disposal or reuse of water. Proper treatment not only reduces environmental impact but also promotes sustainability through resource recovery and efficient water management.

UNIT-3: Other Environmental Pollution & Waste Management

1. Soil Pollution

Soil is a fundamental natural resource that supports plant growth, agriculture, and ecological balance. It is composed of minerals, organic matter, water, and air, forming a dynamic system essential for life. The quality of soil largely depends on its water characteristics, such as moisture content, permeability, and nutrient availability, which directly influence plant productivity. However, increasing human activities have led to soil pollution, which refers to the contamination of soil by harmful chemicals and waste materials, reducing its fertility and usability.

Soil pollution is mainly caused by the excessive use of chemical fertilizers and pesticides, industrial waste disposal, plastic accumulation, and sewage discharge. These pollutants degrade soil quality and introduce toxic substances into the food chain. As a result, soil loses its natural fertility, biodiversity is reduced, and agricultural productivity declines. Moreover, contaminated soil can affect human health through consumption of polluted crops.

Key points:

- Soil contains minerals, organic matter, water, and air
- Soil water affects nutrient availability and plant growth
- Pollution caused by chemicals, industrial waste, plastics
- Leads to loss of fertility and biodiversity
- Causes food contamination and health risks

2. Noise Pollution

Noise pollution refers to unwanted or excessive sound that disrupts normal activities and affects human health. With rapid urbanization and industrial growth, noise levels have increased significantly in cities. Major sources include transportation systems, industrial machinery, construction activities, and public events. Unlike other forms of pollution, noise does not leave visible residues, but its effects are immediate and harmful.

Continuous exposure to high noise levels can lead to hearing loss, stress, sleep disturbances, and reduced concentration. It also affects mental health and can lead to cardiovascular problems. Therefore, controlling noise pollution is essential for maintaining a healthy living environment.

Key points:

- Sources include traffic, industries, construction, loudspeakers
- Causes hearing loss and stress
- Leads to sleep disturbance and reduced efficiency
- Affects mental and physical health
- Controlled by soundproofing, regulations, green belts

3. Nuclear & Radiation Hazards

Radiation pollution occurs due to the release of radioactive substances into the environment from nuclear power plants, medical applications, and nuclear weapon testing. These radioactive materials emit harmful radiations that can penetrate living tissues and cause serious damage. Although nuclear energy is a powerful source of energy, improper handling and accidents can lead to severe environmental consequences.

Exposure to radiation can cause genetic mutations, cancer, and damage to cells and organs. The effects are often long-lasting and may even affect future generations. Proper safety measures, strict regulations, and advanced technologies are required to minimize radiation hazards.

Key points:

- Sources include nuclear plants, medical uses, weapon testing
- Causes cancer, mutations, and tissue damage
- Long-term environmental contamination
- Requires strict safety measures and monitoring
- Safe disposal of radioactive waste is essential

4. Marine Pollution

Marine pollution refers to the contamination of oceans and seas due to human activities such as industrial discharge, oil spills, and excessive plastic waste. One of the major concerns is oil spills, where petroleum products spread over the water surface, blocking oxygen transfer and harming marine organisms. Another serious issue is ocean acidification, caused by increased carbon dioxide (CO₂) absorption by seawater, which lowers the pH and affects marine life, especially corals and shell-forming organisms.

Marine pollution disrupts aquatic ecosystems, reduces biodiversity, and affects human life through the food chain. Since oceans are interconnected, pollution spreads easily and becomes difficult to control, making prevention and regulation extremely important.

Key points:

- Oil spills → blocks oxygen, harms marine life
- Ocean acidification → ↓ pH due to CO₂ absorption
- Causes loss of marine biodiversity
- Pollutants enter food chain via seafood
- Controlled by waste treatment, plastic reduction, regulations

1. Solid Waste Management

Introduction

Solid waste management refers to the systematic process of generation, collection, segregation, treatment, and disposal of solid wastes in a manner that minimizes environmental impact. With rapid urbanization, industrialization, and population growth, the

quantity of waste generated has increased significantly, creating serious environmental and public health issues. Improper waste disposal leads to soil contamination, water pollution, air pollution, and spread of diseases. Therefore, modern waste management focuses on sustainable practices like recycling, reuse, recovery, and eco-friendly disposal techniques.

Composting

Composting is a biological process in which microorganisms decompose organic waste such as food scraps, agricultural residues, and leaves into a stable, humus-like substance called compost. This process occurs under controlled aerobic conditions and converts waste into valuable fertilizer. Composting not only reduces the volume of waste but also improves soil fertility and structure.

Key points:

- Converts biodegradable waste into nutrient-rich compost
- Involves microbial decomposition under aerobic conditions
- Improves soil fertility and water retention
- Reduces burden on landfills
- Eco-friendly and widely used in agriculture

Vermiculture (Vermicomposting)

Vermiculture is an advanced form of composting that uses earthworms to accelerate the decomposition of organic waste. Earthworms consume organic matter and convert it into vermicompost, which is highly rich in nutrients and beneficial microorganisms. This method is faster and produces superior-quality manure compared to traditional composting.

Key points:

- Uses earthworms (e.g., *Eisenia fetida*)
- Produces high-quality organic fertilizer
- Enhances soil microbial activity
- Faster than conventional composting
- Suitable for small-scale and large-scale use

Landfills

Landfills are engineered sites designed for the safe disposal of solid waste by burying it under layers of soil. Modern sanitary landfills include protective measures such as liners, leachate collection systems, and gas recovery units to minimize environmental risks. However, poorly managed landfills can lead to groundwater contamination and methane emissions.

Key points:

- Used for large-scale waste disposal
- Equipped with liners to prevent leakage
- Collects leachate and landfill gases (methane)
- Risk of soil and water contamination if unmanaged
- Requires proper monitoring and maintenance

Hazardous Waste Treatment

Hazardous wastes are those that are toxic, corrosive, flammable, or reactive in nature. These wastes originate from industries, hospitals, and laboratories. Due to their dangerous nature, they require specialized treatment methods to neutralize or destroy harmful components before disposal.

Key points:

- Includes chemical, biomedical, and industrial waste
- Methods: incineration, chemical neutralization, stabilization
- Requires strict safety measures
- Prevents toxic exposure and environmental damage
- Safe disposal in secure landfills

Bioremediation Technologies

Bioremediation is a biological method that uses microorganisms such as bacteria and fungi to degrade or detoxify pollutants present in soil and water. These organisms break down complex organic pollutants into simpler, non-toxic substances. It is widely used for treating oil spills, contaminated soils, and industrial waste.

Key points:

- Uses microorganisms for pollutant degradation
- Effective for organic pollutants and oil spills
- Can be in-situ (on-site) or ex-situ (off-site)
- Eco-friendly and cost-effective
- Converts pollutants into harmless products

Conventional Techniques & Phytoremediation

Conventional Techniques

Conventional remediation techniques are widely used methods for treating contaminated soil and wastewater using natural or semi-engineered processes. These techniques are generally cost-effective, eco-friendly, and suitable for large-scale applications. Among them, land farming and constructed wetlands are important approaches that rely on natural biological processes to degrade pollutants.

Land farming involves spreading contaminated soil over a large area and periodically tilling or ploughing it to enhance aeration and microbial activity. Microorganisms present in the soil break down organic pollutants into simpler and less harmful substances. This method is particularly effective for treating petroleum-contaminated soils and industrial waste.

Constructed wetlands are engineered ecosystems that mimic natural wetlands. They use a combination of plants, soil, and microorganisms to treat wastewater through processes such as filtration, sedimentation, and biological degradation. Water flows through these systems where pollutants are naturally removed, making it an efficient and sustainable treatment method.

Key points:

- Land farming → enhances microbial degradation of pollutants
- Constructed wetlands → natural wastewater treatment system
- Uses plants, microbes, and soil interactions
- Low-cost and environmentally friendly
- Suitable for large-scale and long-term applications

Phytoremediation

Phytoremediation is an advanced green technology that utilizes plants to clean up contaminated soil, water, and air. Certain plants have the ability to absorb, accumulate, or detoxify pollutants such as heavy metals, pesticides, and organic compounds. These plants either store contaminants in their tissues or transform them into less harmful forms.

This method is widely used due to its simplicity, cost-effectiveness, and minimal environmental disturbance. It also improves soil quality and supports ecological restoration. However, it may require more time compared to other treatment techniques.

Key points:

- Uses plants to absorb and degrade pollutants
- Effective for heavy metals and toxic substances
- Improves soil fertility and structure
- Cost-effective and eco-friendly method
- Suitable for large contaminated areas

2. Degradation of Xenobiotics in Environment

Introduction

Xenobiotics are synthetic or man-made chemical substances that are not naturally present in the environment. These include petroleum hydrocarbons, pesticides, plastics, and heavy metals. Due to their complex chemical structure, they are resistant to natural degradation and tend to persist for long periods, causing environmental pollution and health hazards. Their degradation mainly depends on physical, chemical, and biological processes, particularly microbial activity.

Petroleum Hydrocarbons

Petroleum hydrocarbons are released into the environment through oil spills, leakage of fuels, and industrial discharges. These compounds contaminate soil and water and are toxic to living organisms. However, certain microorganisms can degrade hydrocarbons by using them as a source of energy, converting them into simpler compounds like carbon dioxide and water.

Key points:

- Sources: oil spills, refineries, fuel leakage

- Degraded by microbial action (biodegradation)
- Causes toxicity to aquatic organisms
- Reduces oxygen levels in water bodies

Pesticides

Pesticides are widely used in agriculture to control pests but often persist in the environment due to their stable chemical nature. They accumulate in soil and water and enter the food chain, leading to bio magnification. Their slow degradation poses long-term risks to human health and ecosystems.

Key points:

- Persistent and non-biodegradable chemicals
- Cause bioaccumulation and bio magnification
- Affect nervous system and overall health
- Degraded slowly by microorganisms and sunlight

Heavy Metals

Heavy metals such as lead, mercury, cadmium, and arsenic are highly toxic and non-biodegradable pollutants. Once released into the environment, they accumulate in soil and water and enter living organisms through the food chain. Unlike organic pollutants, they cannot be degraded and must be removed or immobilized.

Key points:

- Non-biodegradable and highly toxic
- Accumulate in soil, water, and organisms
- Cause neurological damage and organ failure
- Removed using phytoremediation and chemical treatments

E-Waste Management

Introduction

E-waste (electronic waste) refers to discarded electrical and electronic devices such as computers, televisions, mobile phones, printers, and batteries that have reached the end of their useful life. With rapid technological advancement, frequent upgrades, and reduced product lifespan, the generation of e-waste has increased drastically across the world. This has become a serious environmental concern because e-waste contains both valuable materials and hazardous substances.

Improper disposal methods such as open dumping, burning or informal recycling release toxic chemicals into the environment. These substances contaminate soil, water, and air, posing serious risks to human health and ecosystems. Therefore, scientific management and proper recycling of e-waste are essential for sustainable development.

Key points:

- Rapid increase due to technology growth and consumer demand
- Contains both valuable metals and hazardous substances
- Improper disposal leads to serious pollution and health risks
- Requires systematic management and recycling

Types of E-Waste

E-waste includes a wide range of electronic products from domestic, commercial, and industrial sectors. These wastes vary in composition and require different handling and recycling techniques. Classification of e-waste helps in efficient collection and treatment.

E-waste is broadly categorized into large household appliances, small electronic devices, IT and communication equipment, and electronic components.

Key points:

- **Household appliances** → TV, refrigerator, washing machine, air conditioner
- **IT and communication devices** → computers, laptops, printers, servers
- **Consumer electronics** → mobile phones, tablets, audio/video equipment
- **Electrical tools & components** → circuit boards, wiring, batteries

Environmental Impact

E-waste contains toxic materials such as lead, mercury, cadmium, arsenic, and brominated flame retardants, which are highly harmful to the environment and human health. When e-waste is dumped in landfills, these toxic substances leach into soil and groundwater. When burned, they release hazardous gases into the atmosphere.

Long-term exposure to these pollutants can cause serious health issues such as cancer, neurological disorders, kidney damage, and respiratory problems. In addition, e-waste contributes to ecological imbalance by harming plants, animals, and microorganisms.

Key points:

- Causes soil and groundwater contamination due to leaching
- Burning releases toxic gases and air pollutants
- Leads to serious health problems (cancer, organ damage)
- Affects ecosystems and biodiversity
- Results in long-term environmental degradation

E-Waste Recycling

E-waste recycling is the process of collecting, dismantling, and processing discarded electronic products to recover valuable materials such as metals, plastics, and glass. Proper recycling reduces the need for mining new raw materials and prevents environmental pollution.

The recycling process involves several steps including collection, segregation, dismantling, material recovery, and safe disposal of non-recyclable components. Advanced recycling techniques ensure efficient recovery of precious metals like gold, silver, and copper.

Key points:

- Recovery of valuable metals (gold, copper, silver, aluminium)
- Reduces dependence on natural resources
- Prevents environmental pollution
- Supports circular economy and sustainability
- Requires authorized recycling facilities

E-Waste Management Rules

To control the growing problem of e-waste, governments have introduced strict regulations for its proper handling and disposal. In India, e-waste management is governed by rules that emphasize responsibility, accountability, and safe processing of electronic waste.

One of the key concepts is Extended Producer Responsibility (EPR), where manufacturers are responsible for collecting and recycling their products after use. Proper collection systems, authorized recycling units, and awareness programs are also essential components of these rules.

Key points:

- Extended Producer Responsibility (EPR) → producers must manage end-of-life products
- Establishment of collection centres and take-back systems
- Use of authorized and certified recycling units
- Safe handling of hazardous materials
- Promotion of public awareness and compliance
- Strict monitoring and implementation by authorities.